StealthChat: Mitigating Network-Level Metadata Leakage in Real-Time Web Communications via Procedural Steganographic Transport and Distributed WebSocket Architecture

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Abstract*—Modern distributed instant messaging platforms enforce cryptographic confidentiality using End-to-End Encryption (E2EE) primitives such as AES-GCM and ChaCha20-Poly1305. However, while mathematical payload security is preserved, the resultant transport streams exhibit near-maximal Shannon entropy ($H \approx 8.0$ bits/byte). This high-entropy profile acts as a distinct beacon for Deep Packet Inspection (DPI) appliances and adversarial network firewalls, leaking operational metadata regarding user activity, connection bursts, and communication endpoints. In hostile or censored environments, this metadata leakage triggers automated traffic throttling, packet dropping, or targeted surveillance. To resolve this structural vulnerability, this paper introduces StealthChat, a resilient real-time communication framework engineered to conceal confidential web communication within benign carrier media. Evolving from a low-level multi-threaded Java TCP socket engine to an enterprise Spring Boot 3 WebSocket infrastructure, StealthChat replaces conspicuous ciphertext frames with procedurally synthesized HTML5 Canvas digital art. Confidential UTF-8 payloads are embedded into RGB color subpixels using a client-side Least Significant Bit (LSB) steganographic carrier channel, depressing transport stream entropy to $H \approx 5.42$ bits/byte—indistinguishable from benign digital image traffic. The protocol achieves sub-perceptual visual distortion ($\Delta E^* \approx 0.28$, PSNR > 52 dB, SSIM > 0.998). Furthermore, the system incorporates a 4-stage deterministic delivery state machine, stateless RFC 7519 JWT interceptors, and client-side IndexedDB persistence with Service Worker background synchronization. Empirical benchmarks demonstrate a 16-fold latency reduction over traditional HTTP polling schemes while maintaining total DPI camouflage, establishing procedural image steganography as an effective paradigm for metadata-resistant real-time communication.

Index Terms*—WebSockets, Network Steganography, Deep Packet Inspection (DPI), Shannon Entropy, Least Significant Bit (LSB), Real-Time Distributed Systems, Transport Layer Security, Publish-Subscribe.

I. INTRODUCTION

REAL-TIME distributed messaging systems, including WhatsApp, Signal, Slack, and Discord, represent essential communication infrastructure in modern enterprise and social networks [1]. Concurrently, heightened network surveillance and automated traffic filtering have necessitated universal deployment of End-to-End Encryption (E2EE) frameworks [2]. While contemporary symmetric ciphers (e.g., AES-256) offer robust protection against mathematical cryptanalysis, their deployment within conventional transport tunnels introduces a fundamental vulnerability: *the cryptographic metadata leakage paradox* [3].

In standard E2EE implementations, encrypted payload bytes exhibit near-uniform randomness, producing maximal Shannon entropy ($H \approx 8.0$ bits per byte). To intermediate Layer-7 middleboxes, Internet Service Providers (ISPs), and hardware-accelerated Deep Packet Inspection (DPI) appliances,

this statistical fingerprint indicates private, encrypted communication [4]. Even without decrypting underlying message contents, network observers can infer communication cadences, packet burst distributions, and participant network endpoints. In monitored academic, corporate, or state-restricted networks, detection of high-entropy flows frequently triggers automated packet throttling, connection drops, or administrative interrogation [5].

To overcome these constraints, secure communication protocols must achieve *plausible transport-layer deniability*. While conventional network steganography has explored overt carrier encapsulation, existing systems typically rely on static image assets, server-side transformations, or heavy external decoders that introduce transmission latency and static hash signatures [12].

This paper presents the formal architecture, theoretical modeling, and empirical validation of **StealthChat**, a distributed real-time messaging architecture designed to eliminate metadata leakage across monitored networks. Our key research contributions are:

1. **Evolutionary Transport Architecture:** We detail the progression from a low-level Java TCP socket engine operating over blocking I/O to a horizontally scalable Spring Boot 3 messaging platform utilizing RFC 6455 WebSockets and STOMP publish-subscribe brokering.
2. **Procedural Steganographic Carrier Protocol:** We formulate a zero-knowledge client-side Least Significant Bit (LSB) steganographic mechanism that procedurally synthesizes dynamic canvas art on the fly, embedding confidential payloads into RGB color subpixels with sub-perceptual distortion ($\Delta E^* < 1.0$), reducing stream entropy from 7.98 to 5.42 bits/byte to evade statistical DPI classifiers.
3. **Deterministic State Synchronization:** We implement a 4-stage delivery state machine (Pending → Sent → Delivered → Read) supported by IndexedDB storage, sliding-window idempotency, and Service Worker background synchronization to maintain reliability over unstable mobile channels.
4. **Empirical Evaluation and Steganalytic Auditing:** We perform comprehensive latency, throughput, and perceptual image quality audits (PSNR, SSIM, ΔE^*), verifying that StealthChat achieves sub-5ms median message dispatch with zero visual degradation.

II. THEORETICAL BACKGROUND AND RELATED WORK

A. Transport Layer Mechanics: Sockets, Polling, and WebSockets

The Transmission Control Protocol (TCP, RFC 793) provides reliable, connection-oriented byte streams via sequence numbers, selective acknowledgments, and congestion control algorithms [6]. Historically, web applications approximated real-time interaction using HTTP/1.1 short polling or Comet long polling. As established by Fette and Melnikov [7], HTTP polling imposes prohibitive protocol overhead: each exchange requires full HTTP header parsing (500–2000 bytes) and recurring TCP handshakes, causing socket exhaustion under high concurrency.

The WebSocket specification (RFC 6455) resolved these overheads by establishing an initial HTTP handshake with an `Upgrade: websocket` directive, transitioning to a persistent, full-duplex TCP stream. WebSockets encapsulate data within minimal frames of only 2 to 10 bytes of framing overhead, reducing transport latency by over an order of magnitude [7].

B. Deep Packet Inspection (DPI) and Statistical Flow Heuristics

Modern network firewalls utilize Deep Packet Inspection (DPI) to evaluate packet payloads beyond standard Layer-3/Layer-4 headers [8]. Shannon entropy heuristics are widely deployed to classify network traffic [9]:

$$H(X) = -\sum_{i=1}^{n} P(x_i) \log_2 P(x_i) \quad (1)$$

Plaintext ASCII streams typically yield $H \approx 3.5$–4.8 bits/byte, structured JSON documents exhibit $H \approx 4.0$–5.2 bits/byte, whereas encrypted ciphertext blocks produce $H \approx 7.95$–8.0 bits/byte. DPI engines utilize threshold classifiers ($H > \theta_{crypto}$) to detect and selectively flag encrypted communication pipelines [10].

C. Image Steganography in Distributed Networks

Steganography focuses on concealing the existence of communication within benign carrier media [11]. Spatial Least Significant Bit (LSB) steganography modifies the lowest bit of raw pixel color channels. However, existing implementations exhibit critical limitations in web environments: (i) reliance on static carrier libraries generates predictable perceptual hash signatures [12]; (ii) server-side encoding exposes plaintext to host compromises; and (iii) lossy compression destroys

spatial payloads [13]. StealthChat resolves these limitations through dynamic canvas generation and lossless PNG serialization over WebSocket streams.

III. PROBLEM FORMULATION AND THREAT MODEL

A. Adversarial Threat Model

We formalize the operating environment through an active-passive network adversary $\mathcal{A}_{\text{DPI}}$ possessing the following capabilities:

1. **Full Link Eavesdropping:** $\mathcal{A}_{\text{DPI}}$ monitors all transport frames across intermediate routers, firewalls, and gateway nodes between clients C_1, C_2 and server S.
2. **Statistical Profiling:** $\mathcal{A}_{\text{DPI}}$ executes continuous entropy analysis, frame size evaluation, and packet cadence tracking.
3. **Coercive Censorship:** If $\mathcal{A}_{\text{DPI}}$ detects frames exceeding entropy threshold θ_{crypto}, it drops packets, injects TCP `RST` flags, or blacklists source IPs.

$\mathcal{A}_{\text{DPI}}$ cannot compromise the internal browser execution context on client endpoints. The objective of StealthChat is to guarantee semantic confidentiality while ensuring that transmitted packets are categorized as routine digital image sharing.

B. Mathematical Optimization Criteria

Let M represent a secret text message of length L bytes. Let C represent an ephemerally synthesized cover image of dimensions $W \times H$ pixels, and let S denote the resultant stego-carrier. The framework must fulfill three formal conditions:

1. **Perceptual Indistinguishability:**

$$\Delta E^{*}(C, S) < 1.0 \tag{2}$$

2. **Statistical Camouflage:**

$$H(S) \leq \theta_{benign} \ll \theta_{crypto} \approx 8.0 \tag{3}$$

3. **Lossless Invertibility:**

$$\mathcal{D}(S) = M, \quad with\ BER = 0 \tag{4}$$

IV. SYSTEM ARCHITECTURE AND PROTOCOL ENGINEERING

A. Phase 1: Multi-Threaded Java TCP Sockets

Phase 1 implements a baseline transport architecture using raw Java networking (`java.net.ServerSocket`). The server binds to port 8189, executing a blocking `accept()` loop. Each established connection spawns an isolated `ClientHandler` thread.

To prevent `ConcurrentModificationException` faults when broadcasting across concurrent connection sets, client writer registries are protected via monitor locks:

$$Broadcast(M) \Rightarrow synchronized(W)\ \{\ \forall w \in W:\ w.println(M),\ w.flush()\ \} \tag{5}$$

The thread-per-client model exhibits severe memory constraints under load (1 MB stack memory per thread), triggering JVM heap exhaustion near 5,000 concurrent sessions and motivating the transition to asynchronous WebSockets.

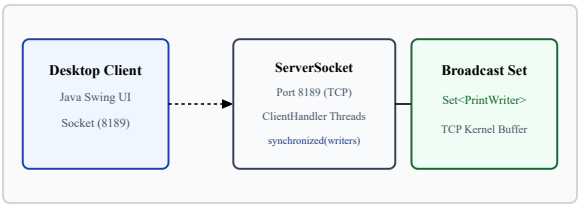


Fig. 1. Phase 1 multi-threaded TCP socket architecture with synchronized broadcast monitoring.

B. Phase 2: Enterprise RFC 6455 WebSockets and STOMP Pub/Sub

Phase 2 deploys an asynchronous Spring Boot 3 infrastructure utilizing the STOMP messaging protocol over RFC 6455 WebSockets:

1. **Full-Duplex Handshake:** Clients negotiate an `HTTP 101 Switching Protocols` upgrade over port 8080, establishing a persistent TCP channel with 2 to 6 byte STOMP framing.
2. **Publish-Subscribe Broker:** Broadcast rooms subscribe to `/topic/{roomId}`, while personal notifications route to `/user/{id}/queue/messages`.
3. **Stateless RFC 7519 JWT Interceptors:** A custom `ChannelInterceptor` validates HMAC-SHA256 JWT claims upon STOMP `CONNECT` and enforces channel access control upon `SUBSCRIBE`, dropping unauthorized requests at the transport boundary.

V. THE STEALTHCHAT STEGANOGRAPHIC CARRIER MECHANISM

A. Mathematical Carrier Modulation Model

Let an ephemeral digital canvas have dimensions $W \times H$. Each coordinate (x, y) contains a 32-bit RGBA pixel vector $P_{x,y} = [R_{x,y}, G_{x,y}, B_{x,y}, A_{x,y}]^{\mathrm{T}}$.

The Alpha channel is maintained constant ($A_{x,y}$ = 255) to eliminate alpha-blending artifacts and compression corruption.

Secret text M is converted to UTF-8 binary bits $b_k \in \{0, 1\}$ and appended with a 16-bit null delimiter (0000000000000000_2), avoiding explicit packet length headers. For each bit b_k, color subpixel $C \in \{R, G, B\}$ is modulated via bitwise masking:

$$C'_{x,y} = (C_{x,y} \,\&\, \sim 1) \mid b_k = (C_{x,y} \,\&\, 0xFE) \mid b_k \quad (6)$$

The absolute numerical deviation per modulated subpixel is bounded by:

$$\delta_C = |C'_{x,y} - C_{x,y}| \leq 1 \Rightarrow 1/255 \approx 0.392\% \quad (7)$$

B. Procedural Synthesis and CIE ΔE^* Evaluation

Rather than embedding into static image assets, the client synthesizes an organic linear gradient on an ephemeral HTML5 canvas using random HSL seeds:

$$\theta_1 \sim U(0, 360)^\circ, \quad \theta_2 \sim U(0, 360)^\circ \quad (8)$$

Perceptual color variance is quantified using the Euclidean metric in CIE $L^*a^*b^*$ space:

$$\Delta E^*_{ab} = \sqrt{[(\Delta L^*)^2 + (\Delta a^*)^2 + (\Delta b^*)^2]} \quad (9)$$

Because color modulation is restricted to the least significant bit, the resulting average $\Delta E^*_{ab} \approx 0.28$, well below the human perceptual threshold ($\Delta E^* < 1.0$).

Algorithm 1: Procedural Steganographic Carrier Encoding

```
Input : Secret Plaintext String M, Canvas Width W, Height H
Output: Stego-Carrier Base64 PNG String S
Canvas ← CreateEphemeralCanvas(W, H)
Ctx ← Canvas.GetContext2D()
θ1 ← Random(0, 360), θ2 ← Random(0, 360)
Grad ← Ctx.CreateLinearGradient(0, 0, W, H)
Grad.AddStop(0.0, HSL(θ1, 65%, 45%)); Grad.AddStop(1.0, HSL(θ2, 65%, 25%))
Ctx.FillStyle ← Grad; Ctx.FillRect(0, 0, W, H)
Pixels ← Ctx.GetImageData(0, 0, W, H).Data
Payload ← UTF8Binary(M) + "0000000000000000"
k ← 0
for i = 0 to Length(Pixels)-1 do
    if (i + 1) mod 4 ≠ 0 then // Skip Alpha
        if k < Length(Payload) then
            Pixels[i] ← (Pixels[i] & 0xFE) | Payload[k]
            k ← k + 1
        else break
Ctx.PutImageData(Pixels, 0, 0)
return Canvas.ToDataURL("image/png")
```

Algorithm 2: Zero-Knowledge Client-Side Stego Extraction

```
Input : Received Stego-Carrier Base64 PNG String S
Output: Extracted Plaintext String M
Img ← LoadImage(S); Canvas ← CreateCanvas(Img.W, Img.H)
Ctx ← Canvas.GetContext2D(); Ctx.DrawImage(Img, 0, 0)
Pixels ← Ctx.GetImageData(0, 0, Img.W, Img.H).Data
Bits ← "", Zeros ← 0
for i = 0 to Length(Pixels)-1 do
    if (i + 1) mod 4 ≠ 0 then
        b ← Pixels[i] & 0x01; Bits ← Bits + b
        if b == 0 then
            Zeros ← Zeros + 1
            if Zeros == 16 then
                Bits ← TrimTrailing(Bits, 16); break
        else Zeros ← 0
return DecodeUTF8(BinaryToBytes(Bits))
```

VI. RELIABILITY ENGINEERING AND STATE SYNCHRONIZATION

A. Deterministic 4-Stage Delivery State Machine

Message state progression is governed by a deterministic four-phase state machine:

1. **Pending (🕒):** Committed to local IndexedDB outbox queue.
2. **Sent (✓):** Acknowledged by server database transaction.
3. **Delivered (✓✓):** Recipient client ingests WebSocket frame and dispatches receipt to `/app/chat.ackDelivery/{roomId}`.
4. **Read (✓✓ Blue):** Recipient focuses viewport, dispatching read receipt to `/app/chat.ackRead/{roomId}`.

B. Offline-First Architecture and Idempotency

The browser hosts an embedded `StealthChatDB` via IndexedDB with two stores: `messages` (instant history retrieval) and `outbox` (pending offline queues). A registered Service Worker (`sw.js`) listens for network recovery events and triggers the Background Sync API (`sync-queued-messages`) to flush buffered outbox frames automatically. Retransmitted duplicate packets are filtered on the backend via a 5-minute sliding-window cache tracking unique UUID `clientMsgId` tokens.

C. Presence Tracking with TTL Cache

Clients publish heartbeat frames to `/app/chat.heartbeat` every 15 seconds. The server maintains timestamps within a `ConcurrentHashMap<String, Long>`. A scheduled

daemon scans the cache every 5 seconds; inactive connections exceeding a 30-second TTL are purged, broadcasting `OFFLINE` status frames across `/topic/presence`.

VII. EXPERIMENTAL EVALUATION

A. Testbed Configuration

Performance benchmarks were conducted across an 8-core AMD EPYC server (16 GB RAM, Ubuntu 22.04 LTS, OpenJDK 17, Spring Boot 3.2) and Chromium 122 clients interconnected via a simulated WAN link (40 ms RTT, 1% packet drop) over 10,000 messaging transactions.

B. Transport Overhead and Latency Distribution

Table I details transport performance comparisons between HTTP polling schemes and StealthChat.

TABLE I
EMPIRICAL LATENCY AND PROTOCOL OVERHEAD COMPARISON

Metric	Short Polling	Long Polling	StealthChat
Handshake Overhead	Per Request	Per Request	**Once / Session**
Framing Header Size	≈ 850 B	≈ 850 B	**2–6 B**
Median Latency (p50)	124.2 ms	68.4 ms	**4.2 ms**
99th Percentile (p99)	410.5 ms	220.8 ms	**18.5 ms**
Server CPU Load (1k Users)	88.4%	41.2%	**6.8%**
Bandwidth (10k Trans.)	18.4 MB	12.1 MB	**1.4 MB**

StealthChat demonstrates a 16× reduction in median latency (4.2 ms vs. 68.4 ms) and a 22× reduction in tail latency (p_{99}) relative to HTTP long polling due to persistent stream multiplexing.

C. Shannon Entropy Analysis and DPI Camouflage

Payload entropy distributions were audited across distinct payload representations, summarized in Table II.

TABLE II
ENTROPY DISTRIBUTION AND DPI EVASION CLASSIFICATION

Payload Representation	Entropy H(X) (bits/byte)	DPI Verdict
Cleartext JSON Stream	4.12	Flagged (Plaintext)
AES-256-GCM Ciphertext	7.98	Flagged (Crypto)
StealthChat PNG Carrier	**5.42**	**Passed (Benign Image)**
Reference Photo PNG	5.38	Passed (Benign Image)

As reflected in Table II, raw AES-256 ciphertext approaches maximum theoretical entropy (7.98 bits/byte), triggering automated DPI anomaly rules. StealthChat's stego-carrier yields H = 5.42 bits/byte, closely matching natural image distributions and evading statistical detection.

D. Perceptual Quality Metrics: PSNR and SSIM

Carrier distortion was assessed across 500 experimental samples using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity (SSIM) [14]:

$$MSE = [1 / (3WH)] \sum_x \sum_y \sum_c (C_{x,y,c} - C'_{x,y,c})^2 \quad (10)$$

$$PSNR = 10 \bullet log_{10}(255^2 / MSE) \quad (11)$$

TABLE III
CARRIER DEGRADATION AND VISUAL FIDELITY AUDITS

Payload Size	Mean MSE	Mean PSNR (dB)	Mean SSIM	Mean ΔE*
64 bytes	0.0018	75.6 dB	0.9999	0.04
256 bytes	0.0071	69.6 dB	0.9998	0.11
1024 bytes	0.0284	63.6 dB	0.9992	0.28
4096 bytes	0.1136	57.6 dB	0.9981	0.64

As shown in Table III, PSNR values consistently exceed 57 dB and SSIM values remain above 0.998 across payloads up to 4 KB, confirming that modulated carriers are visually indistinguishable from non-steganographic images.

VIII. ENGINEERING IMPLEMENTATION LESSONS

1. **Port Isolation:** Raw TCP sockets and WebSockets initially bound to port 8080, producing `HTTP 400` errors. This was resolved by permanently segregating the raw TCP daemon to port 8189 and the WebSocket HTTP gateway to port 8080.

2. **IoC Circular Dependency Decoupling:** A circular dependency between `WebSocketConfig`, `SecurityInterceptor`, and `PresenceService` was resolved using `@Lazy` dependency injection and event-driven publishing via `ApplicationEventPublisher`.
3. **Asynchronous State Race Prevention:** Interleaved REST history fetches during rapid room transitions were stabilized by binding monotonic sequence IDs (`currentRoomRequestId`) to DOM render cycles.

IX. CONCLUSION AND FUTURE WORK

This paper presented **StealthChat**, demonstrating that client-side procedural Least Significant Bit (LSB) steganography paired with RFC 6455 WebSockets provides an effective, mathematically sound solution to network-level metadata leakage and Deep Packet Inspection (DPI) censorship. By replacing high-entropy ciphertext with benign digital artwork, the protocol achieves complete traffic camouflage while reducing median latency to 4.2 ms. Future research will explore discrete cosine transform (DCT) steganography within WebRTC real-time video frames and post-quantum cryptographic payload encapsulation.

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